

PAPER_284: M16 Eagle Nebula UQFF — Dual Mass Co-Action Product (Φ_{dm})

SFR Growth × Photoevaporation Erosion Multiplicative Coupling

Classification: UQFF 2.0 Gravitational Physics — Nebular Mass Dynamics

System: M16 Eagle Nebula (IC 4703), Eagle Nebula Star-Forming Region

Session: 80 | **Module:** M16_UQFF_MODULE.cpp (22nd C++ UQFF module)

Author: Daniel T. Murphy | **Date:** March 2026

Abstract

This paper introduces the **Dual Mass Co-Action Product** (Φ_{dm}) — a UQFF gravity modulation factor that couples star-formation-driven mass accumulation and radiation-driven photoevaporation erosion through a **multiplicative** product rather than the previously used additive form. For M16 (Eagle Nebula), with star formation rate $SFR = 1 \text{ M}\odot/\text{yr}$ over initial gas mass $M_0 = 1200 \text{ M}\odot$, and maximum photoevaporation fraction $E_0 = 0.3$ (30%), the multiplicative form produces a 24.3% reduction in Φ_{dm} relative to the additive approximation at $t = 5 \text{ Myr}$. This is the **first UQFF module** to simultaneously apply an additive-gain and saturation-subtractive product on the same gravity term.

2. Physical Motivation

In active star-forming nebulae, two competing processes drive mass evolution:

1. **Star Formation Accretion** — molecular gas accretes onto protostars, increasing the effective gravitational mass fraction by $SFR_rate \times t$.
2. **Photoevaporation Erosion** — UV radiation from newly formed massive stars erodes the surrounding gas, progressively reducing the effective mass by a saturating fraction $E_{rad}(t)$.

The **additive approximation** (used in prior UQFF modules):

$$\Phi_{dm}^{add} = g_{base} \times (1 + M_{sf}) - E_{rad}$$

linearly superposes the two effects, implicitly treating them as independent processes acting on separate mass reservoirs.

The **multiplicative form** (this paper):

$$\Phi_{dm}(t) = (1 + SFR_rate \times t) \times (1 - E_{rad}(t))$$

correctly encodes that **the eroded mass is drawn from the same growing reservoir** — the fraction lost to photoevaporation scales with the mass being accreted, not the original quiescent mass. This is physically accurate for pillar-geometry star formation (e.g., M16's "Pillars of Creation").

3. Mathematical Formulation

3.1 Parameters (M16 Eagle Nebula)

Parameter	Value	Description
M_0	$2.387 \times 10^{33} \text{ kg (1200 } M_\odot)$	Initial nebula gas mass
SFR	$1 M_\odot/\text{yr}$	Star formation rate
SFR_rate	$2.639 \times 10^{-11} \text{ s}^{-1}$	$= \text{SFR} / (M_0/M_\odot) / (3.156 \times 10^7 \text{ s/yr})$
τ_{erode}	$9.468 \times 10^{13} \text{ s (3 Myr)}$	Photoevaporation e-folding time
E_0	0.3	Maximum photoevaporation fraction
g_{base}	$1.454 \times 10^{-12} \text{ m/s}^2$	$G \times M / r^2$ at $r = 3.31 \times 10^{17} \text{ m}$

3.2 Dual Co-Action Product

$$M_{sf}(t) = \text{SFR_rate} \times t$$

$$E_{rad}(t) = E_0 (1 - e^{-t/\tau})$$

$$\Phi_{dm}(t) = (1 + M_{sf}) \times (1 - E_{rad})$$

3.3 Dynamic Gravity Term

$$g_{dyn}(t) = g_{base} \times \Phi_{dm}(t)$$

3.4 Multiplicative-Additive Gap

The gap relative to the additive form:

$$\Delta_{gap} = \Phi_{dm}^{mult} - \Phi_{dm}^{add} = -(M_{sf} \times E_{rad})$$

This cross-term is always **negative** — the multiplicative form predicts lower gravity than the additive approximation whenever both SFR accumulation and erosion are simultaneously active.

4. Numerical Results at t = 5 Myr

$$t = 5 \text{ Myr} = 1.578 \times 10^{14} \text{ s}$$

Quantity	Value
M_sf_frac	$\text{SFR_rate} \times t = 2.639 \times 10^{-11} \times 1.578 \times 10^{14} = \mathbf{4164.8}$
E_rad	$E_0 \times (1 - \exp(-5/3)) = 0.3 \times 0.8110 = \mathbf{0.2433}$
Φ_{dm} (multiplicative)	$(1 + 4164.8) \times (1 - 0.2433) = 4165.8 \times 0.7567 = \mathbf{3151.9}$
Φ_{dm} (additive)	$(1 + 4164.8) - 0.2433 = \mathbf{4165.6}$
gap_mult_add	$-(4164.8 \times 0.2433) = \mathbf{-1013.3}$ (24.3% less)
$g_{dyn}(5 \text{ Myr})$	$1.454 \times 10^{-12} \times 3151.9 = \mathbf{4.583 \times 10^{-9} \text{ m/s}^2}$

The multiplicative gap of -1013.3 confirms that treating erosion as acting on the growing mass reservoir (not the static initial mass) produces a **measurable 24.3% reduction** compared to the additive approximation.

5. Connection to UQFF 2.0 g_{total}

In the full M16 UQFF 2.0 equation:

$$g_{\text{total}}(r, t) = [g_{\text{dyn}}(t) + U_{g,\text{sum}}(26) + \Lambda + Q + L + F + g_{\text{exp}}] \times \text{corr}_{SC}$$

The Φ_{dm} product modulates only the dynamic base gravity term g_{dyn} . The 26-layer Triadic ($U_{g,\text{sum}}$), cosmological Λ , quantum, Lorentz, fluid, and Friedmann expansion terms are all independent of Φ_{dm} — the modulation is cleanly scoped to the time-evolving mass component.

6. UQFF Historical Distinction

Module	SFR Term	Erosion Term	Form
Session 55 CP3 M16EagleNebulaRadiationSFR	$g_{\text{base}} \times (1 + M_{\text{sf}})$	$-E_{\text{rad}}$	Additive
This paper (PAPER_284)	$(1 + M_{\text{sf}})$	$\times (1 - E_{\text{rad}})$	Multiplicative

This is the **first UQFF module** to use the multiplicative dual co-action form, correctly encoding the coupled feedback between star formation accretion and pillar photoevaporation for M16-class nebulae.

7. Wolfram KB Term

M16UQFF:Phi_dm=(1+SFR_rate*t)*(1-E_0*(1-Exp[-t/tau])); SFR_rate=2.639e-11/s; M_sf_frac=SFR_rate*t [PAPER_284]

8. Cross-References

- **PAPER_285:** Erosion Saturation Half-Time (t_{half} , Δg_{Max})
 - **PAPER_286:** Nebular Friedmann Redshift (κ_{neb} , $z=0.0015$)
 - **M16_UQFF_MODULE.cpp:** Full UQFF 2.0 C++ implementation (22nd module)
 - **CondensedPhysics3.py:** M16DualMassCoActionProductCalculator
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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **nebula-formation** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{neb}})(\partial^\mu \phi_{\text{neb}}) - V(\phi_{\text{neb}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{neb}}) = \frac{1}{2}m^2\phi_{\text{neb}}^2 + \frac{\lambda}{4!}\phi_{\text{neb}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{neb}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{neb}}} = \nabla \cdot (\rho_{\text{neb}} \nabla \phi) + \rho_{\text{vac},[\text{SCm}]} \cdot (P_{\text{rad}}/c) = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{neb}} =$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.126 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 47, \quad n_{\text{channel}} = 25/26$$

Since $p_{\text{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10⁶ yr** (Jeans collapse timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.126	✓ Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 47$	✓ Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26,$ $\cos(2\pi j/26)$	✓ Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	✓ Canonical
[SSq]	0.57	Applied in BSH saturation	✓ Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] * n / 26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}} / F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma^2)] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Ly_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms

Module	Purpose	Key Result
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q;q)_{n^2}$ - $\phi_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q^{2;q^2})_n$ - $\psi_{26}(q) = \sum_{n=1}^{26} q^{\{n2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	8-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-kappa_i * n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_Scm	0.99	SCm manifold completeness
rho_Scm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_Scm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_1_CoAnQi.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl, uqff_mock_theta_pi_kernel.wl).